KRITICAL HACKS 2026 | AI/ML TRACK · STUDENT CATEGORY | DEMO VIDEO ↗ | LINKEDIN POST ↗ | GITHUB ↗

JAAGRUK

जागरूक Janta's AI Agent for Government Rights, Updates & Knowledge

One profile.
Everything the government owes you.

An AI agent that reads the live government web through TinyFish, then finds, verifies and helps citizens claim schemes, scholarships, exams, insurance and unclaimed money, in their own language.

Agentic AI | Neuro-symbolic ML | Built on TinyFish | GovTech

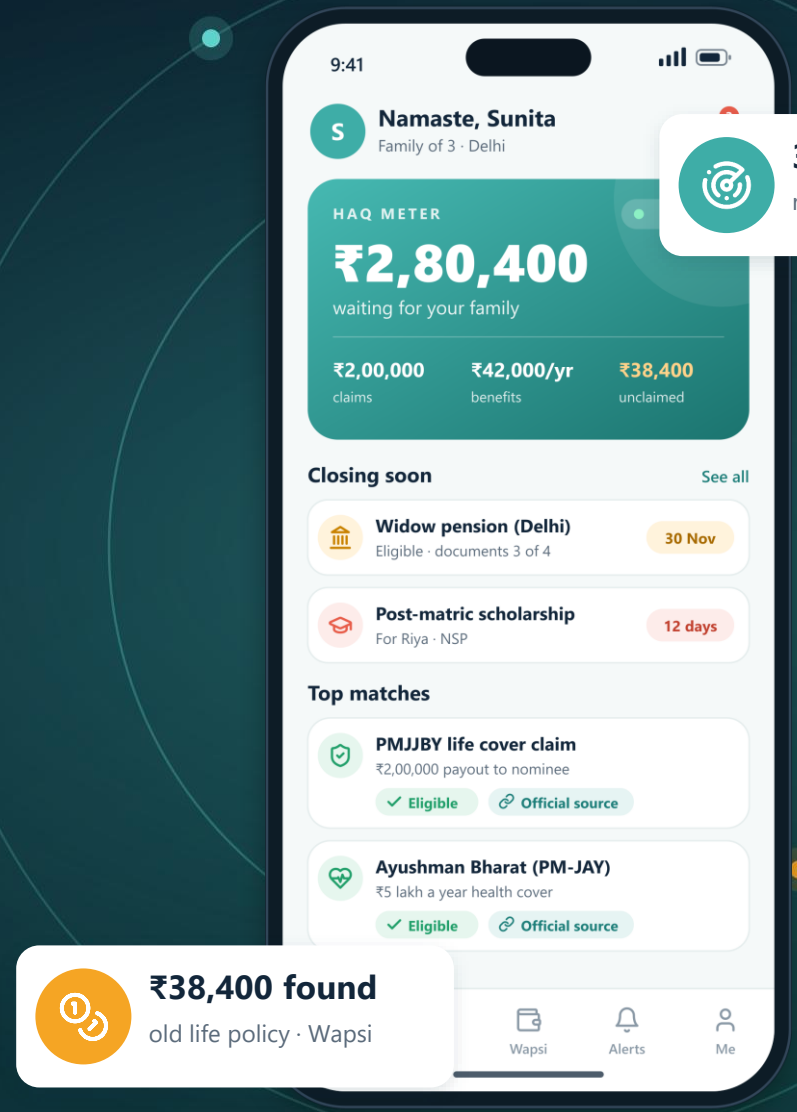
THE TEAM

SH Sagar Hundia — Full-stack & app

AA Ayush Awasthi — LLM & Scam Shield

SN Saif Nasim — Agents & ML lead

YK Yash Kumar — Backend & rules



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Benefits exist. Citizens never find them.

₹98,073 Cr

of citizens' money sits unclaimed in RBI's Depositor Education & Awareness Fund

Finance Ministry data, Jan 2026

12%

of older Indians know the Annapurna food-security scheme exists

India Ageing Report 2023, UNFPA–IIPS

₹3,216 Cr

lost to fake job & investment scams in 11 months (1 lakh+ cases)

Govt data, Jan–Nov 2024

100s

of portals and PDF notice boards, each in its own format, with no single live feed

Central + 36 States/UTs + PSUs

WHY CITIZENS MISS OUT

Fragmented
Central, 36 States/UTs and PSUs each run separate portals

Unstructured
Eligibility rules buried in scanned PDFs and legal language

Changes silently
Deadlines extended, corrigenda issued, nobody is alerted

Not built for citizens
English-first, literacy-heavy and never personalised

Exploited by scammers
Fake scheme and job sites fill the information gap

OUR PROBLEM STATEMENT

Millions of eligible Indians miss schemes, scholarships, jobs, insurance and even their own money because information is fragmented across hundreds of portals, buried in unstructured notices, changes without warning and is rarely in their language. **Portals list information; nobody watches, matches, verifies and acts for the individual citizen.**

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JAAGRUK watches, matches, verifies and acts for you

WHAT'S DIFFERENT

- Personal, not generic
- Live, not static
- Verified, not guessed
- Acts, not just lists
- In your language

01 Haq Meter
One number for the family: ₹ of benefits, claims and unclaimed money it is missing

02 Life-event mode
Baby born, Class 12 result, job loss, death in family: an instant entitlement checklist

03 Wapsi
Finds forgotten deposits, insurance, shares, dividends and PF across official sources



04 Scam Shield
Checks whether a scheme, job offer or website is real before anyone pays

05 Change Radar
Alerts only when a deadline, rule or vacancy that affects you changes

06 Acts for you
Pre-fills forms, tracks status, drafts grievances; you approve every step

1. Profile in 90 seconds
2. Agents read portals live
3. AI verifies eligibility
4. You claim, on time

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One app, personalised for every kind of citizen

31.89 Cr unorganised workers registered on e-Shram (Aug 2026)

4.5 Cr students in higher education (AISHE 2023-24)

14.9 Cr Indians aged 60+, rising to 34.7 Cr by 2050

Students
Scholarships, entrance exams, fake-internship check

Job aspirants
Exams you qualify for, age-relaxation calculator, date forecasts

Farmers
PM-KISAN status, crop-insurance deadlines, best mandi price

Gig & daily-wage workers
e-Shram benefits, pension and insurance, ration portability

Self-employed & MSMEs
Mudra and PMEGP loans, tenders, compliance radar

Private employees
Old PF accounts found, ESIC and insurance, housing subsidy

Government employees
Pay and DA circulars, departmental exams, NPS vs UPS

Women & homemakers
Maternity, girl-child and self-help-group schemes

Senior citizens
Pensions, Ayushman Vay Vandana (70+), dormant accounts

Families & legal heirs
Death-in-family checklist and Wapsi claim kits

HOW WE PERSONALISE

Language · **State** · **Age** · **Occupation** · **Income** · **Life events**

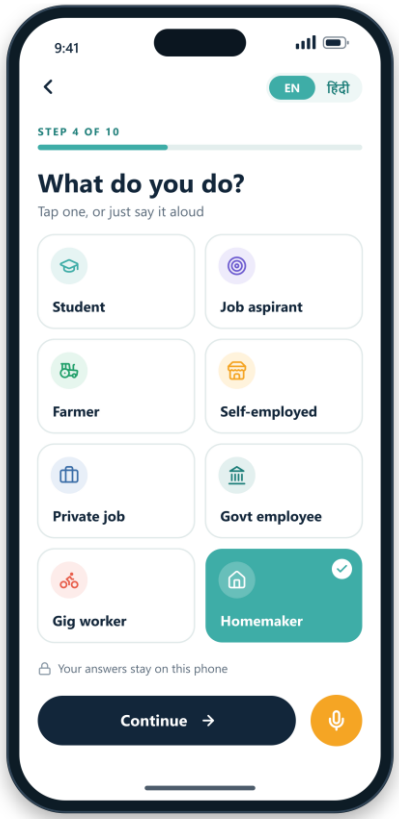
10 voice-friendly questions build a multi-label persona:
e.g. Student + Woman + SC + Rural

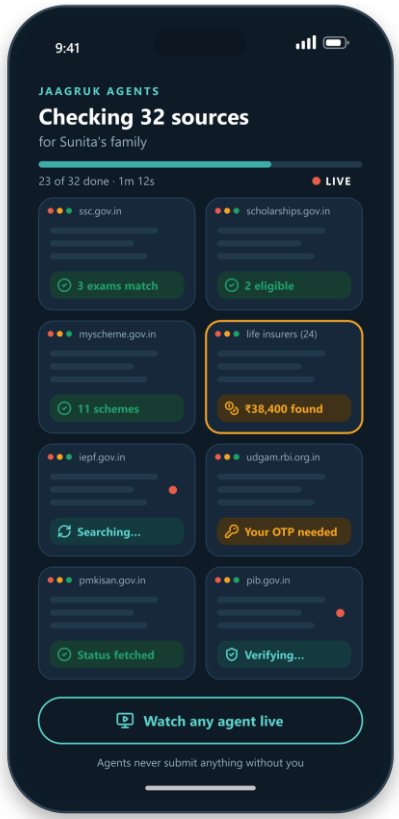
ALSO SERVES (B2B2C · B2G)

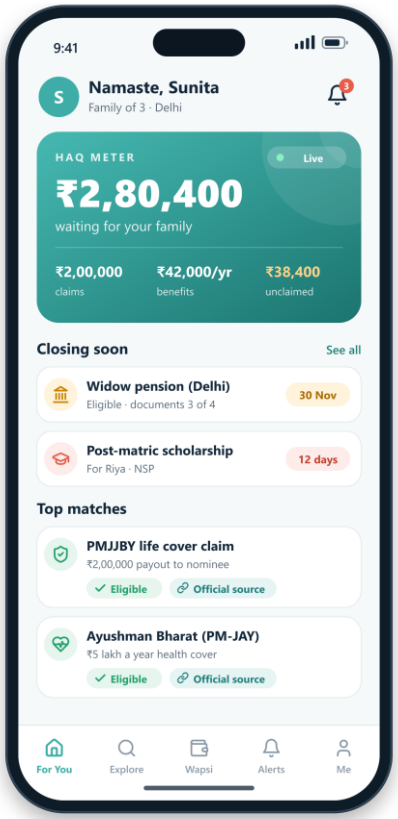
- **CSC & NGO operators**
- **College scholarship cells**
- **District administrations**
- **Banks & insurers (API)**

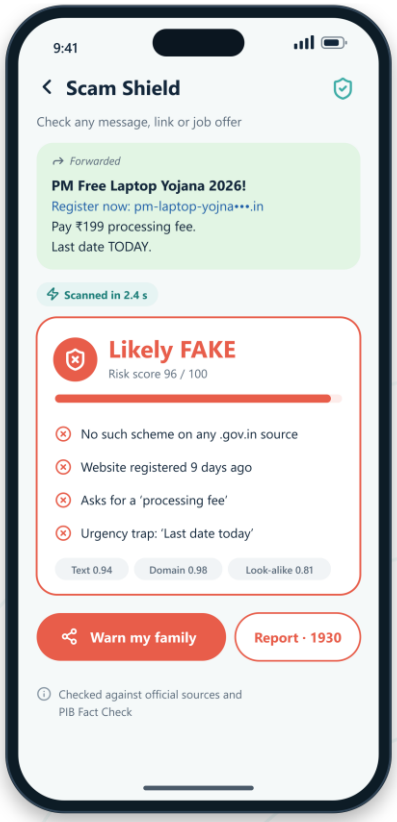
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Use case: Sunita finds ₹2.8 lakh of entitlements in one sitting









1 She speaks, it listens
Sunita, 52, answers 10 questions by voice, in Hindi

2 Agents read 32 portals
TinyFish agents search insurers, IEPF, NSP live

3 AI verifies & ranks
₹2,80,400 waiting: claim, pension, scholarship

4 Scam Shield guards her
A fake 'free laptop' forward is flagged

SAME FLOW, EVERY PERSONA Student: 3 exams + 2 scholarships · Farmer: PM-KISAN + crop-insurance deadline · Gig worker: e-Shram benefits

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Six AI models turn messy web data into verified answers

Rule extraction

Indic OCR · LayoutLMv3 · Claude

PROBLEM	Eligibility hides in scanned, bilingual PDFs
HOW	OCR + layout model; LLM fills a strict JSON schema
OUTPUT	**Machine-readable rules, each linked to its source clause**

Neuro-symbolic eligibility

LLM rules → Z3 SMT solver

PROBLEM	An LLM alone can hallucinate who is eligible
HOW	Rules compiled to logic; Z3 proves yes/no
OUTPUT	**Exact decisions + 'get 1 document, unlock ₹48,000'**

Name matching (Wapsi)

IndicXlit · Siamese network

PROBLEM	Records say 'R K Sharma' or 'Ramesh Sarma'
HOW	Transliterate, then a contrastive model scores pairs
OUTPUT	**Calibrated match: '92% likely your father's policy'**

Scam detection

MuRIL · XGBoost · CLIP ensemble

PROBLEM	Fake sites copy official logos and Hindi text
HOW	Text, domain and look-alike models vote together
OUTPUT	**Risk score with human-readable reasons**

Grounded voice assistant

Bhashini · LaBSE RAG · DeBERTa-NLI

PROBLEM	Questions come in Hinglish; answers can't be made up
HOW	Cross-lingual retrieval; NLI checks every sentence
OUTPUT	**Answers in Indian languages, every line cited**

Ranking & forecasting

LambdaMART · bandits · Prophet

PROBLEM	60 matches, but only 5 fit on a phone screen
HOW	Learning-to-rank + alert bandits + date forecasts
OUTPUT	**Ranked feed, smart alerts, 'notice expected in April'**

TRAINING & EVALUATION DATA, COLLECTED BY TINYFISH AGENTS

Official notices
gold set for rule extraction (field F1)

PIB Fact Check debunks
scam labels vs real schemes (F1)

Public record formats
name pairs for matching (precision)

Past notices & cutoffs
time series for forecasting (MAE)

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An agentic + neuro-symbolic pipeline, built on TinyFish

Government web	Powered by TinyFish · TinyFish harness	Understand	Trust & reason	Deliver
Exam portals: UPSC, SSC, PSCs	Planner picks sources per user	Indic OCR + LayoutLMv3	Z3 solver: exact eligibility	App + WhatsApp bot
Scholarship & scheme portals	Parallel agents on 4 TinyFish APIs	LLM → eligibility rules (JSON logic)	Unlock-path optimiser	Voice in Indian languages (Bhashini)
Insurers · IEPF · RBI UDGAM	Verifier: cross-source check	Multilingual embeddings	RAG + NLI fact-check	Alerts & deadline calendar
PDF notices · PIB · gazettes	Self-healing re-plan	Knowledge graph (Neo4j)	Scam & name-match models	CSC & district dashboards

HOW ONE REQUEST FLOWS

1 Profile
Voice or tap onboarding; profile stays on the phone

2 Plan
Planner picks only the portals that matter for this profile

3 Read
TinyFish agents run in parallel, streamed live

4 Decide
AI extracts rules; Z3 decides eligibility

5 Deliver
Ranked answers with sources; alerts scheduled

TECH STACK

- **Frontend** — React Native · Next.js PWA
- **Backend** — FastAPI · Celery + Redis
- **Web agents** — TinyFish Python SDK
- **LLM** — Claude Sonnet / Haiku
- **Indic AI** — Bhashini · AI4Bharat models
- **Reasoning** — Z3 SMT solver · Neo4j
- **Data** — PostgreSQL + pgvector · S3
- **Privacy** — On-device ONNX · DPDP-ready

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Government data has no API. TinyFish is our engine.

WHY TINYFISH

7 of 9 source groups in our MVP plan have no public API

- ✕ Exam notices: UPSC, SSC, PSCs
- ✕ NSP & state scholarship portals
- ✕ State scheme portals & gazettes
- ✕ Insurers' unclaimed-amount pages
- ✕ IEPF & company dividend lists
- ✕ RBI UDGAM (OTP login)
- ✕ PIB Fact Check
- ○ myScheme (API Setu, approval)
- ✓ data.gov.in · PIB press RSS

No API? TinyFish agents read the site the way a person would.

TINYFISH CAPABILITIES → JAAGRUK FEATURES

Search API
`recency_minutes`
Discovers new schemes, recruitment notices and fake-scheme reports every day

Fetch API
`ttl: 0`
Turns notice pages and PDFs into clean text for the rule-extraction model

Web Agent
`output_schema`
Fills eligibility checkers and unclaimed-money forms on many sites in parallel

Browser + Profiles
`use_profile`
Consented logins for status checks; the citizen types the OTP

Monitor (add-on)
`webhook`
Page & topic monitors push deadline or rule changes to Change Radar

agents/exam_watch.py

```python
from tinyfish import TinyFish
tf = TinyFish()  # TINYFISH_API_KEY
run = tf.agent.run(
  url="https://ssc.gov.in",
  goal=("List new exam notices: "
        "name, last date, age limit, "
        "fee, PDF link. JSON only."),
  output_schema=EXAM_SCHEMA,
)
# run-sse: STREAMING_URL → live grid
```

RESPONSE SHAPE

```
{ "status": "COMPLETED",
  "result": { "exams": [ {
    "name", "last_date", "age_limit", … } ] } }
```

WHAT WE WILL DEMO LIVE

1. Search + Fetch read a real notice
2. 10+ Web Agents run in parallel, streamed
3. Browser session with user-typed OTP

Planner — picks sources for the profile
→ **TinyFish agents** — run in parallel, streamed live
→ **Verifier** — COMPLETED ≠ success: checks result
→ **Self-heal** — re-plans from the goal on failure
→ **Cache + diff** — only real changes become alerts

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Entitlements reach the citizens they were meant for

₹98 Cr

returned to families for every 0.1% of India's ₹98,073 Cr unclaimed-deposit pool that Wapsi helps recover

0.1% × ₹98,073 Cr

₹32 Cr

kept with families if Scam Shield stops just 1% of the ₹3,216 Cr lost to job & investment scams

1% × ₹3,216 Cr (Jan–Nov 2024)

Days → mins

to find every relevant scheme, exam and deadline, instead of days of portal-hopping

Pilot target

WHO BENEFITS, AND HOW

Citizens

Discover and claim benefits they never knew existed

Students & job-seekers

Never miss a scholarship, exam or deadline

Seniors & families

Pensions, health cover and unclaimed money recovered

Government

Higher scheme uptake; anonymised district-level gap data

Society

Fewer scam victims and more trust in digital services

KPIs WE WILL MEASURE IN THE PILOT

- Entitlements surfaced per family
- Applications completed before deadline
- Alert precision (useful vs noisy)
- ₹ unlocked: benefits + unclaimed money
- Scam messages flagged
- Users served in Indian languages

UN SDGs ADVANCED



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Free for every citizen. Paid by the institutions that gain

WHO PAYS, HOW MUCH, AND WHY THE PRICE IS FAIR

Citizens

₹0

free, forever

Who: Every family using the app

Why it's fair: Genuine schemes never charge a fee, so neither do we. Free use builds trust and reach.

CSC & NGO console

₹199

per centre / month

Who: 5.16 lakh Common Service Centres, NGOs, college cells

Why it's fair: A CSC earns ₹30 per scholarship application. 7 extra applications a month pay for it.

Banks & insurers

1%

of settled claim, capped at ₹500

Who: Banks, life insurers, mutual-fund AMCs

Why it's fair: RBI pays banks 5–7.5% per settled deposit. On a ₹25k claim we take ≈ ₹250, only on success.

Districts (B2G)

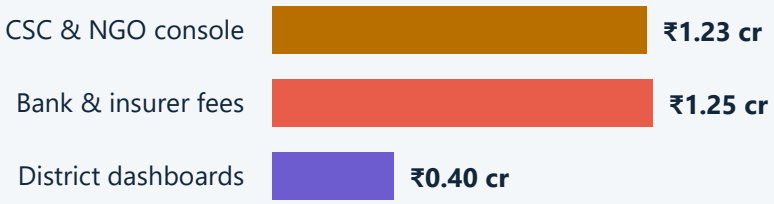
₹2 lakh

per district / year

Who: District administrations, welfare departments

Why it's fair: Free 3-month pilot, then a licence for "eligible but not enrolled" gap maps.

YEAR-2 PROJECTION (CONSERVATIVE)



≈ ₹2.9 cr

a year, with zero revenue from citizens

Assumes 1% of CSCs · 50,000 settled claims × ≈ ₹250 · 20 districts

UNIT ECONOMICS (MEASURED LIVE)

- **≈ ₹8.5** TinyFish cost per full Wapsi search
- **₹0** for Search, Fetch, Z3 proofs, Haq Meter and Scam Shield
- **≈ ₹180/mo** for Change Radar monitors, shared by every user
- **Profitable** at year-2 scale without charging a single citizen

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Buildable in 24 hours, on proven APIs and open models

OUR 24-HOUR BUILD PLAN

H0–3	H3–8	H8–13	H13–18	H18–21	H21–24
Setup + TinyFish goals for 15 portals	**Agent harness + Fetch → LLM rule extraction**	**Z3 eligibility engine, Wapsi matcher, Scam Shield**	**App, live agent grid, Hindi voice**	**End-to-end tests on real portals**	**Demo script + buffer**
Saif · Yash	Saif · Ayush	Yash · Ayush	Sagar	All four	All four

Why it's feasible

- Hard parts are APIs: TinyFish (web), Bhashini (voice), Claude (extraction)
- A new portal is one plain-English goal, not a new scraper
- Models are pretrained (MuRIL, LaBSE, DeBERTa); we only fine-tune
- Team has shipped multi-agent LLM apps and real-time backends

Risks → mitigation

1. **CAPTCHA / OTP** → citizen completes it in live view; never bypassed
2. **Portal redesigns** → goal-based agents re-plan; verifier flags failures
3. **Wrong eligibility** → Z3 proof + NLI check + official-source link
4. **Sensitive data** → on-device matching, consent, no Aadhaar stored (DPDP Act)

Low cost to run

- **Search & Fetch:** free on TinyFish
- **Monitor:** $0.005 per run, ≈ $0.15/month for a daily check
- **Agent & Browser:** credits only for user-specific checks
- **Shared cache:** one fetch of a notice serves every user

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From a Delhi pilot to every Indian, in every language

ROADMAP

1 — 2 — 3 — 4

Hackathon MVP · 24 h
15 portals · Central + Delhi · 3 personas

Pilot · 3 months
100 portals · 2 states · WhatsApp bot

Growth · 9 months
300 portals · 10 states · CSC console

Scale · 18 months
500+ portals · all 36 States/UTs · B2G API

WHAT COMES NEXT

DigiLocker auto-fill
Consent-based document pull to pre-fill applications in one tap

Feature-phone access
IVR and missed-call alerts for citizens without smartphones

All 22 scheduled languages
Full voice support through Bhashini for every state

District gap dashboard
Privacy-preserving uptake data so officials target outreach

Claim-appeal copilot
Contest rejected insurance and benefit claims with cited rules

Open scheme-rules dataset
Machine-readable rules for researchers and other civic apps

PARTNERS WE WILL APPROACH Common Service Centres · NGOs & ASHA networks · college scholarship cells · banks & insurers · state departments

TRY IT YOURSELF

See JAAGRUK working

Watch the demo, read the post, or run it yourself in one command. No API key needed.

Demo video

2–3 minute walkthrough of the live app and TinyFish agents (Google Drive)

drive.google.com/.../Demo

LinkedIn post

What JAAGRUK does, who it helps, and what we learned on real portals

lnkd.in/p/dH2bDxY4

GitHub repo

Open source: code, README, tests, TinyFish evidence and recorded real runs

github.com/SaifNasim047/JAAGRUK-AI

```
$ pip install -r requirements.txt  &&  python run.py --replay
```

Team JAAGRUK · Sagar · Ayush · Saif · Yash